
CONTACT INFORMATION	Department of Biodiversity and Conservation Biology, University of the Western Cape, Robert Sobukwe Way, Bellville, 7535, Cape Town, South Africa Email: ojwachj@gmail.com	
PROFESSIONAL EXPERIENCE	Research Fellow	since July 2019
	University of the Western Cape Institute of Biotechnology and Metagenomics collaborations with Biodiversity and Conservation Biology UWC University of KwaZulu-Natal, Microbiology and Genetics Department of Zoology University of Johannesburg, Kingsway Campus Auckland Park 2006 South Africa	
	Global Challenge Research Fund	January 2019 – March 2019
	University of Glasgow, Glasgow, United Kingdom Young Researchers Exchange Program at the Wellcome Center for Integrative and Molecular Parasitology Institute of Infection, Immunity and Inflammation	
	Demonstrator/ Tutorial Fellow	Durban (SA), Aug 2016 – Dec 2018
	University of KwaZulu-Natal	
	Industry	
	GlaxoSmithKline Pharmaceutical	Nairobi (KE), October 2014 – July 2016
	Laboratory analyst, Quality and Compliance Section	
	GlaxoSmithKline Pharmaceutical	Nairobi (KE), Jan 2016 – July 2016
	Microbiology analyst	

EDUCATION	Master of Science MSc Microbiology (<i>Cum Laude</i>) 2016 - 2018 University of KwaZulu-Natal Dissertation: Synthesis, Detection and Quantification of Inulooligosaccharides and Fructooligosaccharides by Extracellular and Intracellular Inulinase and Fructosyltransferase Enzymes Isolated from Coprophilous Fungi. Supervisors: Prof Samson Mukaratirwa Dr Taurai Mutanda Bachelor of Science BSc Microbiology (Distinction) 2009 – 2013 Jomo Kenyatta University of Agriculture and Technology Kenya Certificate of Secondary Education (KCSE) Grade B+ 2004 – 2007 Nairobi School Kenya Certificate of Primary Education (KCPE) 1996 – 2003 Attained A New Light Junior Academy
RESEARCH ARTICLES	Jeff Ojwach*Ajit Kumar, Samson Mukaratirwa, Taurai Mutanda Fructooligosaccharides Synthesized by Fructosyltransferase from an Indigenous Coprophilous <i>Aspergillus niger</i> strain XOBP48 Exhibits Antioxidant Activity. Bioactive Carbohydrates and Dietary Fibre (2020) Jeff Ojwach*Ajit Kumar, Samson Mukaratirwa, Taurai Mutanda Partial Purification and Characterization of an Extracellular Fructosyltransferase Enzyme from <i>Aspergillus niger</i> sp. XOBP48: Implication in Fructooligosaccharide Production. 3 Biotech (2020). Ojwach Jeff*, Ajit Kumar, Mutanda Taurai, and Mukaratirwa Samson Fructosyltransferase and Inulinase Production by Indigenous Coprophilous Fungi for the Biocatalytic Conversion of Sucrose and Inulin into Oligosaccharides. Journal of Biocatalysis and Agricultural Biotechnology (2020).
PRESENTATIONS & PUBLIC OUTREACH	Seminar South African eDNA workshop Stellenbosch (SA), 6th – 9th Nov 2019 Stellenbosch University

Workshop

Wellcome Trust grant writing workshop Glasgow (UK), 15th of
March 2019
Mugdock Country Park, Theatre Room

Symposium

Wellcome Centre for Integrative Parasitology symposium Glasgow (UK),
27 Feb 2019

Poster presentation: **Jeff Ojwach**, Katie Hughes and Andy Waters*
Does Artemisinin Induce the Formation of Stress Granules in *Plasmodium*?

Ojwach J, and Mutanda T Muldersdrift (SA), 4-7 April 2018
Morphotaxonomic and Molecular Identification of Coprophilous Fungi with
Extracellular Fructosyltransferase and Hydrolase Activities for Production
of Biofunctional Prebiotics.
Oral Presentation at a local conference organized by the South African
Society of Microbiology SASM 2018.

Ojwach J, Mukaratirwa and Mutanda Durban (SA), 25 October 2018
Purification, Characterization and Biotechnological Applications of an
Extracellular Fructosyltransferase Enzyme Extracted from an Indigenous
Coprophilous Fungus.
Oral Presentation at the College of Agriculture Engineering and Science
Research Day on T block, Westville Campus.
Awarded First Prize.

Ojwach J and Mutanda T Durban (SA), 26 October 2017
Screening Coprophilous Fungi for Fructosyltransferase and Inulinase
Biocatalysts for Potential Biotechnological Applications.
Poster Presentation at the College of Agriculture Engineering and Science
Research Day.
Awarded First Prize.

SCHOLARSHIPS AND GRANTS

Full Research Grant

National Research Foundation of South Africa

Cape Town (SA), 2019 – present
Fellowship grant at University of the Western Cape, South Africa
Biodiversity and Conservation Biology

Global Challenge Research Fund (GCRF). University of Glasgow, Glasgow (UK), Jan 2019- April 2019.

Full travel scholarship for best performing student to attend high level training in biomedical science at the Wellcome Trust Center for Integrative and Molecular Parasitology, School of Medical Sciences, University of Glasgow, Scotland.

MSc *Cum laude* Durban (SA), April 2019.
University of KwaZulu-Natal, Vice chancellors award of Master of Science *Cum Laude*

CAES award Durban (SA), 25 Oct 2018
First prize in best poster presentation at the College of Agriculture Engineering and Science Research Day at Westville Campus, University of Kwazulu-Natal.

Travel grant April 2018
School of Life Sciences local conference SASM in Mudlorsdrift Pretoria.

TEACHING AND TRAINEES	Teaching and Demonstration July 2016 – Dec 2018 University of KwaZulu-Natal Taught modules Introduction to Viruses and Fungi MICRO 215 Introduction to Microbiology BIM1 120 Mycology Microbiology for Pharmacists MICR 207
TRAININGS	SFU-SANTHE intensive course on interdisciplinary approaches to understanding the global burden of HIV and youth. SANTHE “SUB-SAHARAN AFRICAN NETWORK FOR TB/HIV RESEARCH EXCELLENCE” Doris Duke Medical Research Institute at the University of Natal’s Nelson R. Mandela School of Medicine. Durban (SA), March 2018 – April 2018
SKILLS AND EXPERTISE	Proteins Biochemistry: Protein expression and purification, FPLC (expert in using AKTA purification system for protein purification), HPLC, gel filtration, affinity chromatography, enzymatic assays, enzyme kinetics, Western blot, ELISA electrophoresis and protein folding/unfolding studies. Routine molecular biology procedures

	Immunoprecipitation and Immunofluorescence Cytofixation of parasites, such as <i>Plasmodium falciparum</i>, <i>Leishman</i> and <i>Trypanosomes</i> RNA isolation from cell culture Animal handling, rodents: exposure to allergens DNA extraction using Qiagen DNeasy Kit Sybr safe DNA electrophoresis <i>In vitro</i> drug sensitivity assays with <i>P. berghei</i> COSHH <i>In vitro</i> culture of parasites Fluorescence microscopy Gene cloning, protein trafficking and protein expression Transfection, mutagenesis and gene knockouts Multi-color Flow cytometry, FACS analysis cloning, ligation, transformation and gene expression of proteins Bioinformatics tools (Bio-Linux, BioJava) Data mining and sequence analysis, database similarities searches, multiple sequence alignment, phylogenetic analysis, submitting DNA sequences to the databases, biostatistics, statistical analysis tools PCR, DDGE, SDS-PAGE (Electrophoresis) Biocatalysis, protein purification HPLC, LC-MS MS and GC-MS
MEMBERSHIPS	South African Society for Microbiology (SASM) American Society of Microbiology (ASM)
REGISTERED WITH PROFESSIONAL BODY	South African Council for Natural Scientific Professions.
REFEREES	Prof Samson Mukaratirwa Discipline: Biology (Westville Campus) School of Life Sciences College of Agriculture, Engineering and Sciences University of KwaZulu-Natal Private Bag X54011 Durban, 4001, South Africa E: mukaratirwa@ukzn.ac.za Andy P. Waters, Ph.D., FRSE Professor of Molecular and Developmental Parasitology

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